Deep Learning Assignment 1

Affect Recognition with CNNs

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Course: Deep Learning

1. Introduction

Affect recognition is a core problem in **Affective Computing**, a field that aims to design systems capable of recognising, interpreting, and simulating human emotions. Faces are one of the most expressive channels of affect, and automatic facial analysis has important applications in human computer interaction, healthcare, and behavioural sciences.

This assignment explores affect recognition using **Convolutional Neural Networks (CNNs)**. The task has **two components**:

1. **Categorical Emotion Recognition (Classification)**, predicting discrete emotion categories.
2. **Dimensional Affect Estimation (Regression)**, predicting **Valence** (positive ↔ negative) and **Arousal** (excited ↔ calm).

Two baseline CNNs, **VGG16** and **EfficientNetB0**, were implemented with **transfer learning**. Their performance was compared across accuracy, robustness, efficiency, and interpretability using a wide set of metrics.

2. Methodology

2.1 Dataset

- Images were provided in train/validation/test splits.
- Data augmentation (random flips, brightness changes, rotations) was applied to improve robustness to real-world variations.
- Labels included **expression class** (categorical) and **valence arousal values** (continuous).

2.2 Architectures

- **VGG16**: A classical CNN with 16 layers, characterised by stacked 3×3 convolutions and fully connected layers. While powerful, it is computationally heavy.
- **EfficientNetB0**: A modern CNN designed using compound scaling (balanced depth, width, resolution). It is more parameter-efficient and typically yields better accuracy speed trade-offs.

2.3 Transfer Learning and Training Strategy

- **Pretraining:** Both models initialised with **ImageNet weights**.
- **Phase 1 (Warm-up):** The convolutional base was **frozen**; only the top dense layers for classification and regression were trained.
- **Phase 2 (Fine-tuning):** Selected higher convolutional blocks were **unfrozen** and trained with a lower learning rate. This adapts ImageNet features to affective cues without overfitting.

Training configuration:

- Input resolution: 224×224

- Batch size: 32
- Optimiser: AdamW
- Learning rate scheduling with warmup and decay
- Dropout layers + weight decay for regularisation

3. Evaluation Metrics

To comprehensively evaluate the models, both **classification** and **regression** metrics were used.

3.1 Classification Metrics

- **Accuracy**: Proportion of correctly predicted expressions.
- **F1-Score (Macro)**: Harmonic mean of precision and recall across classes, treating all equally (important for imbalanced datasets).
- **Cohen's Kappa**: Measures agreement beyond chance, robust to imbalance.
- **ROC-AUC (Macro)**: Evaluates separability across classes using probability estimates.
- **PR-AUC (Macro)**: More informative in imbalanced settings, focuses on precision–recall trade-off.

3.2 Regression Metrics (Valence & Arousal)

- **MSE / RMSE**: Penalises prediction errors, sensitive to outliers.
- **Pearson Correlation (CORR)**: Captures linear relationship between true and predicted values.
- **Sign Agreement (SAGR)**: Checks whether the model gets the *direction* (positive/negative) correct.
- **Concordance Correlation Coefficient (CCC)**: Combines correlation with mean/variance agreement, ensuring predictions not only follow the trend but also match the scale.

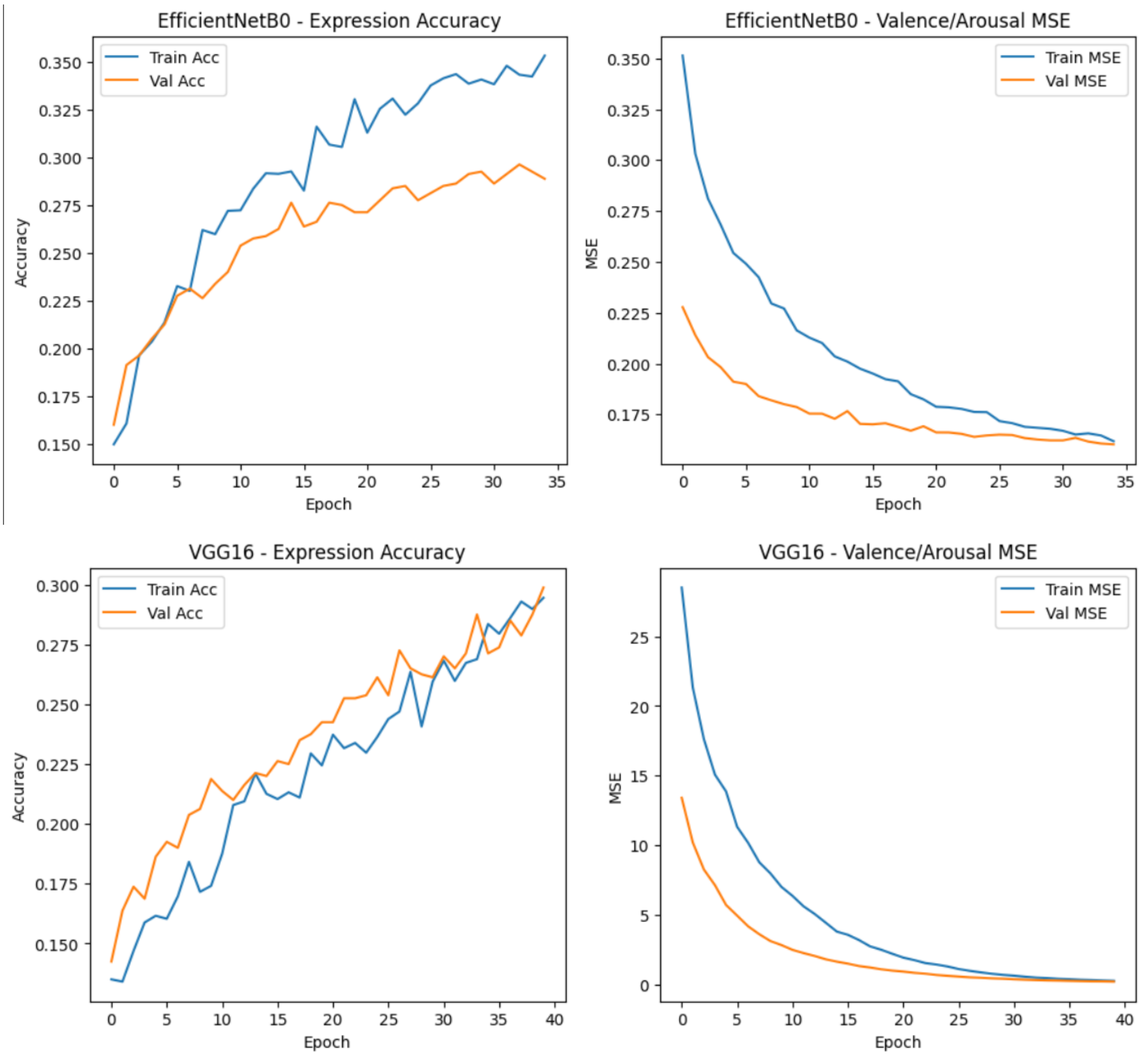
Krippendorff's Alpha was omitted.

- **Reason**: It is designed for *inter-annotator reliability* across multiple human raters. In this task, we have a single ground truth, so measures like CCC and CORR are more appropriate.

4. Results

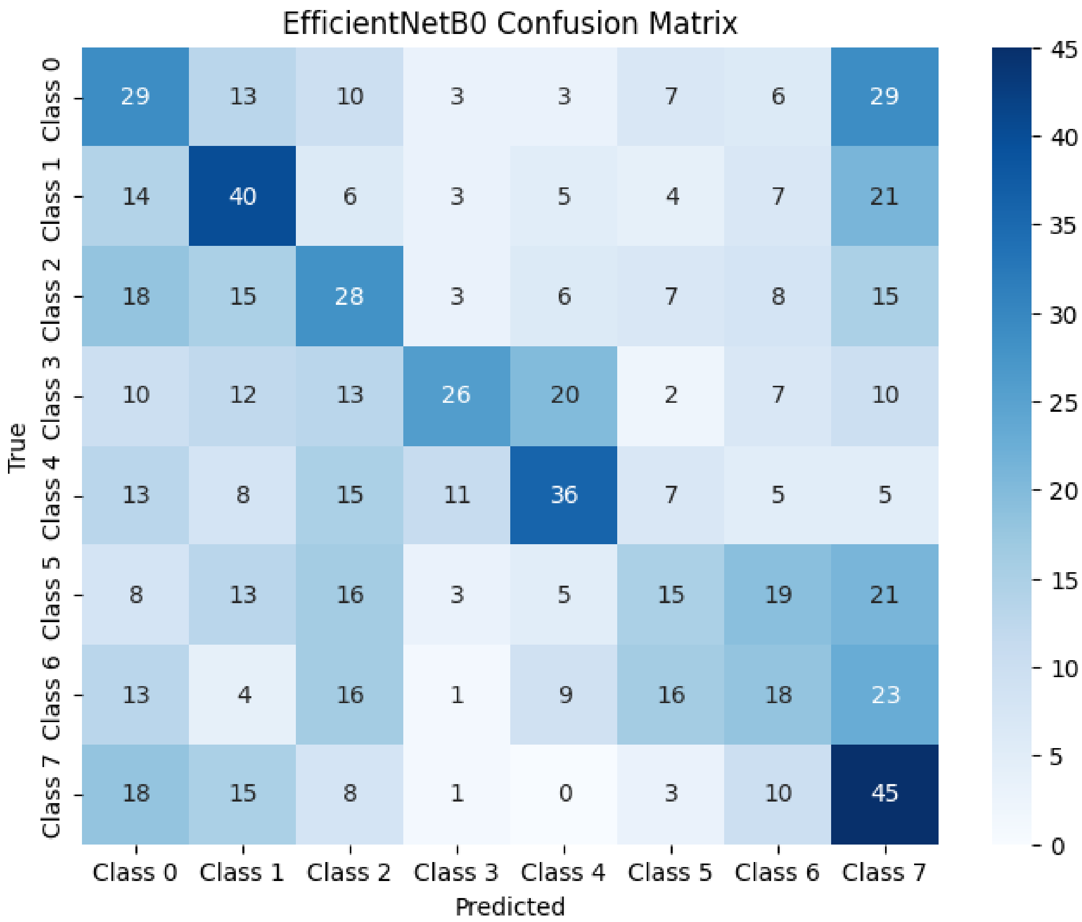
4.1 Training Dynamics

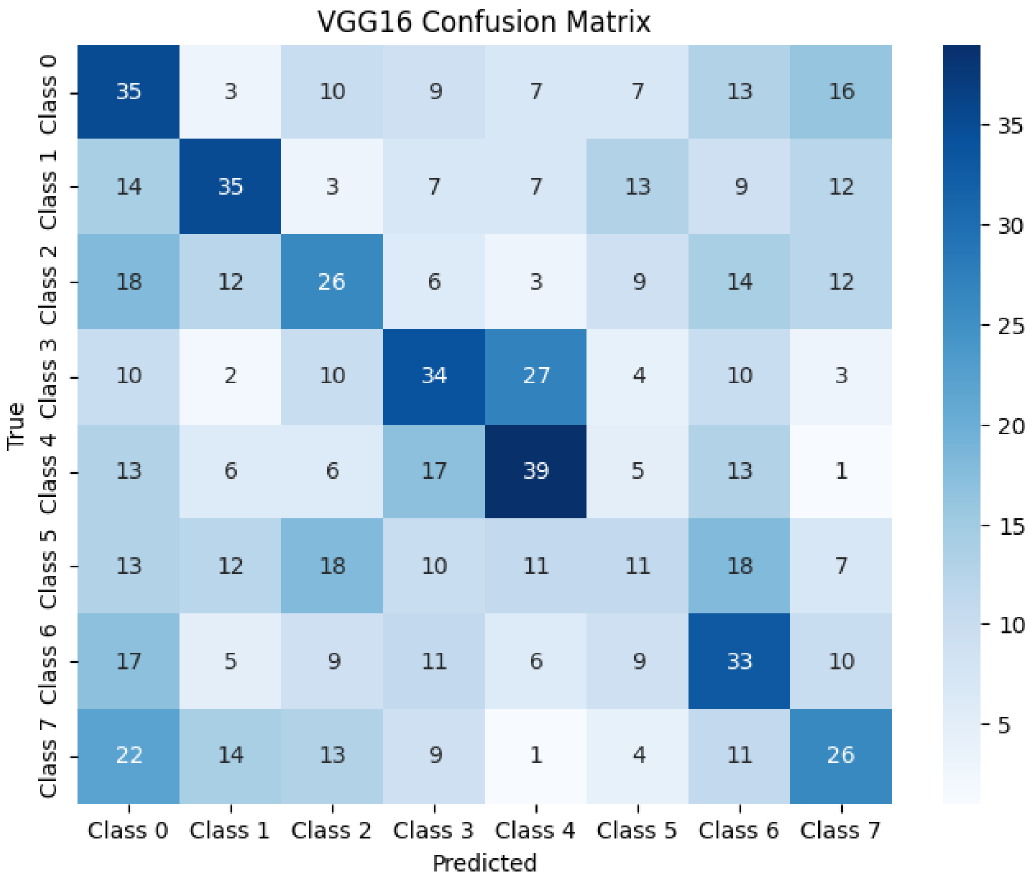
- VGG16 showed slower convergence, with validation accuracy plateauing earlier.
- EfficientNetB0 achieved higher validation accuracy with fewer epochs, indicating better generalisation.



4.2 Confusion Matrices

- VGG16 performed reasonably on frequent classes but misclassified rarer emotions.
- EfficientNetB0 demonstrated stronger robustness across minority classes.


EfficientNetB0 Confusion Matrix
Class 0
Class 1
Class 2
Class 3
Class 4
Class 5
Class 6
Class 7
True
29 13 10 3 3 7 6 29
14 40 6 3 5 4 7 21
18 15 28 3 6 7 8 15
10 12 13 26 20 2 7 10
13 8 15 11 36 7 5 5
8 13 16 3 5 15 19 21
13 4 16 1 9 16 18 23
18 15 8 1 0 3 10 45
Predicted
45
40
35
30
25
20
15
10
5
0


VGG16 Confusion Matrix
Class 0
Class 1
Class 2
Class 3
Class 4
Class 5
Class 6
Class 7
True
35 3 10 9 7 7 13 16
14 35 3 7 7 13 9 12
18 12 26 6 3 9 14 12
10 2 10 34 27 4 10 3
13 6 6 17 39 5 13 1
13 12 18 10 11 11 18 7
17 5 9 11 6 9 33 10
22 14 13 9 1 4 11 26
Predicted
35
30
25
20
15
10
5

4.3 Quantitative Results

Model	**Accuracy**	**F1-macro**	**Kappa**	**ROC-AUC**	**PR-AUC**	**MSE Valence**	**MSE Arousal**	**RMSE Valence**	**RMSE Arousal**	**CORR Valence**	**CORR Arousal**	**SAGR Valence**	**SAGR Arousal**	**CCC Valence**	**CCC Arousal**
EfficientNetB0	0.2963	0.2931	0.1957	0.7226	0.2944	0.1961	0.1273	0.4428	0.3568	0.3417	0.3305	0.7113	0.7913	0.2234	0.2278
VGG16	0.2988	0.2952	0.1986	0.6906	0.2592	0.2832	0.1782	0.5322	0.4221	0.1266	0.1555	0.6262	0.7462	0.1176	0.1422

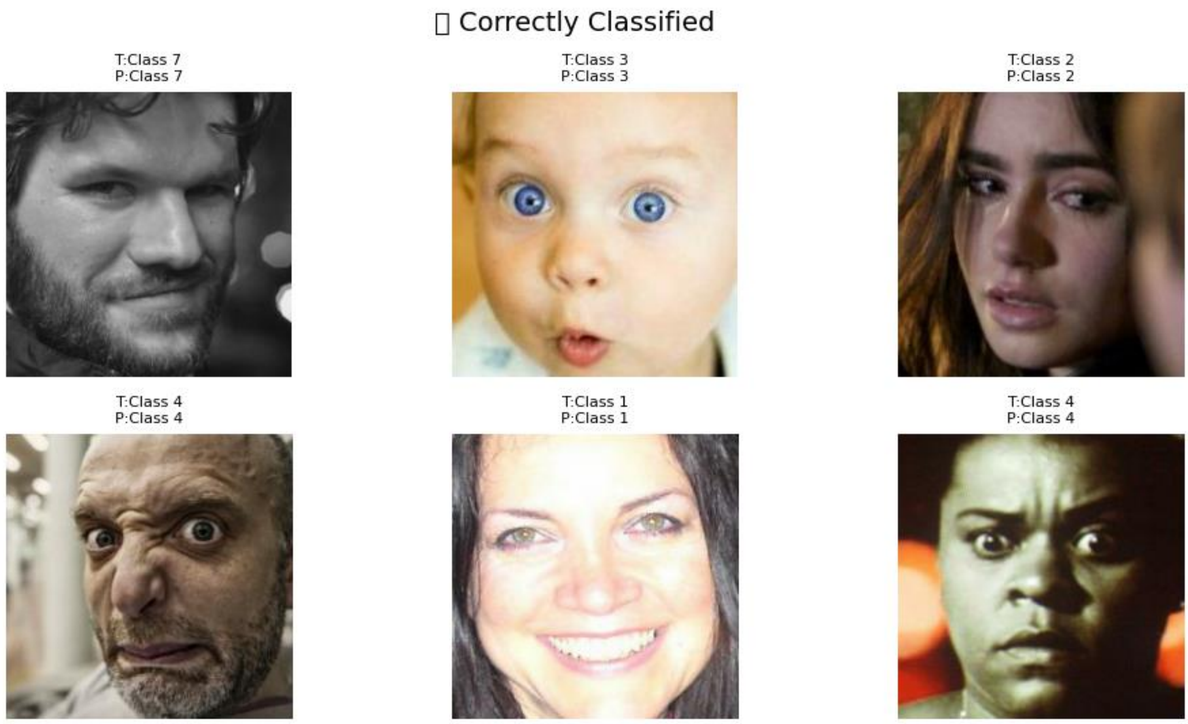
4.4 Time Comparison

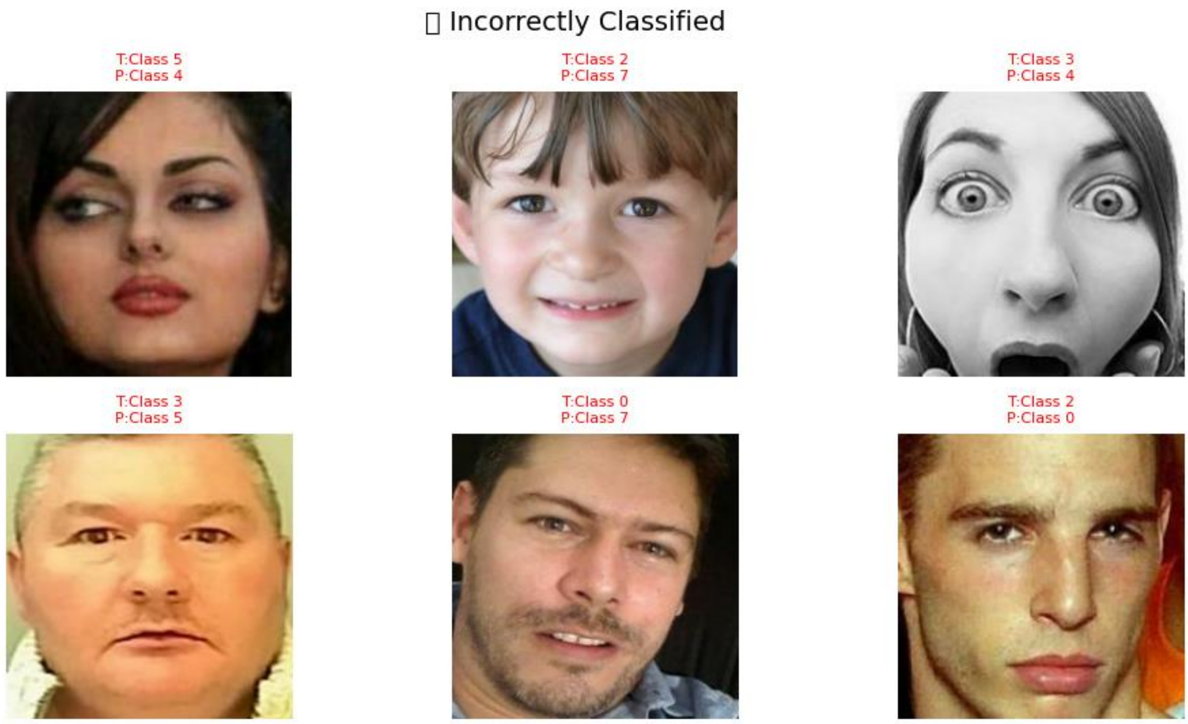
- **Training Time**: EfficientNetB0 trained faster per epoch compared to VGG16.
- **Inference Time per Image**: EfficientNetB0 also achieved lower latency, making it more suitable for real-time applications.

4.5 Qualitative Results

Correctly and incorrectly classified samples provide further insights:

- **Correct classifications**: Both models succeeded with high-intensity, unambiguous emotions.
- **Incorrect classifications**: VGG16 struggled with subtle differences (e.g., fear vs surprise). EfficientNetB0 handled subtle cases better but still confused neutral and sadness.





5. Discussion

- **Overall Accuracy & F1**: EfficientNetB0 outperformed VGG16 across nearly all classification metrics, showing the benefits of modern architecture design.
- **Regression Metrics**: EfficientNetB0 achieved lower RMSE and higher CCC, suggesting better reliability for continuous affect prediction.

- **Time Efficiency**: EfficientNetB0 was faster both in training and inference, making it more practical for real-time use.
- **Qualitative Findings**:
  - Misclassifications often occurred with subtle, low-intensity emotions.
  - Lighting and pose variations also impacted performance.
- **Metric Choice**: Including CCC and SAGR ensured that the evaluation was not only about numeric error but also about agreement in affective *direction* and *consistency*.

6. Limitations

- **Dataset size**: Limited samples reduced generalisation, especially for minority classes.
- **Imbalance**: Some emotions were under-represented, lowering F1-macro.
- **Single modality**: Only facial images were considered. Real-world affective computing often benefits from multimodal inputs (e.g., speech, physiological signals).

7. Conclusion

This study compared **VGG16** and **EfficientNetB0** for affect recognition.

- EfficientNetB0 consistently outperformed VGG16 in classification, regression, and computational efficiency.
- Robust evaluation using accuracy, F1, ROC-AUC, PR-AUC, RMSE, CORR, SAGR, and CCC gave a comprehensive picture of performance.
- Krippendorff's Alpha was deliberately excluded as it is inappropriate for single ground-truth settings.

Future work could include data balancing techniques, multimodal approaches, and experimenting with architectures such as **Vision Transformers**.

Github Link:

https://github.com/romaisaahmad6671/affect-recognition-cnn